KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 26 · SCIENCE

Gravity: Why Things Fall

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 24):** Name the three kinds of plate boundaries. Then explain what happens to an ocean plate at a subduction zone.

2. Why do you think a dropped ball falls down instead of up or sideways? Write your best idea.

3. Predict: On the Moon, would you weigh the same as on Earth? Would your body have the same amount of matter? Explain.

4. Draw Earth as a circle with three people standing in different places around it. Draw an arrow from each person's feet showing which way "down" is.

5. Predict: Does a falling ball move at the same speed the whole way down, or does its speed change? Why?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Measuring the Pull

Use evidence from today's lesson. Write in complete sentences.

1. **Mass or weight?** A 3-kilogram melon: mass on the Moon? Weight on Earth (9.8) and on the Moon (1.6)?

2. **Read the data.** NASA lists Mars at 3.7 and Saturn (cloud level) at 9.0 newtons on each kilogram, though Saturn has about 95 times Earth's mass. Use mass AND distance to explain.

3. **Two-step.** With no air, how fast is a falling object moving after 2 seconds? How much faster after 5?

4. **Find the mistake.** Padma says, "A pan balance would show a smaller mass on the Moon." What is wrong?

★ **Challenge: Explain.** NASA says gravity at the space station is about 90 percent as strong as on the ground. So why do astronauts float?

Looking Ahead: Next week: Earth's Water by the Numbers. How much water is salty, fresh, frozen and underground? You will graph it like a scientist.